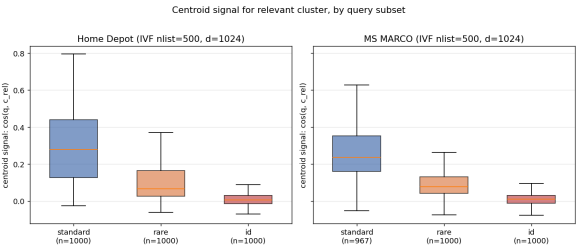


# Direct evidence: measure the centroid signal

**For each query:** compute  $\langle q, c_{\text{rel}} \rangle$  on the centroid of the query's *relevant* cluster (IVF,  $n_{\text{list}}=500$ ,  $d=1024$ , HD).



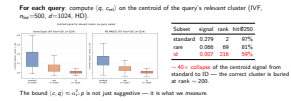
The bound  $\langle c, q \rangle \approx \alpha_{t^*}^2 p$  is not just suggestive — it is what we *measure*.

| Subset   | signal | rank | hit@250 |
|----------|--------|------|---------|
| standard | 0.279  | 2    | 97%     |
| rare     | 0.066  | 69   | 81%     |
| id       | 0.007  | 216  | 59%     |

~ 40× collapse of the centroid signal from standard to ID — the correct cluster is buried at rank ~ 200.

# Beyond Representation: Index Structure Limitations in Dense Retrieval

Direct evidence: measure the centroid signal



**Framing.** This is where we stop arguing from theory and start measuring the exact quantity the bound predicts.

**What we did (per query).** Train IVF on Home Depot ( $n_{\text{list}} = 500$ ,  $d = 1024$ ). For every query, look up which cluster actually contains its relevant doc(s) — call that centroid  $c_{\text{rel}}$ . Then compute  $\langle q, c_{\text{rel}} \rangle$ , and also *where*  $c_{\text{rel}}$  ranks among all 500 centroids scored against  $q$ , and whether it makes the top 250 (“hit@250” — the halfway mark).

**Histogram.** Standard queries: signal ~ 0.28, clearly separated from zero. Rare queries collapse toward zero (~ 0.07). ID queries pile up at ~ 0.007 — **40× smaller than standard**.

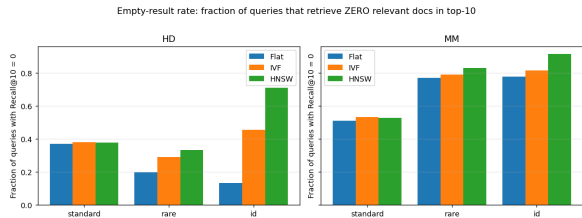
**Table.** Standard: correct centroid at rank 2 on average, hit@250 = 97% — IVF with  $n_{\text{probe}} = 5$  is basically always correct. Rare: rank 69, so you’d need  $n_{\text{probe}} \approx 70$  (huge). **ID: rank 216 out of 500 — worse than median.** hit@250 = 59% means **41% of ID queries have their relevant cluster in the bottom half** of the centroid ranking. Those are structurally unrecoverable short of scanning all clusters — which defeats IVF’s purpose.

**Tie to the bound.**  $\langle c, q \rangle \approx \alpha_{t^*}^2 p$ . Rare tokens: huge  $\alpha$  (IDF), tiny  $p$  (prevalence). Product falls under the JL floor  $\epsilon$  — exactly the collapse we measure.

**Anticipated question.** “Just use finer clusters?” It helps (rare/ID gain 2–3× more relatively than standard going  $n_{\text{list}} 200 \rightarrow 1000$ ) but you’re fighting  $p$ : shrinking clusters means more centroids to score,

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# The production-visible symptom



## Fraction of queries with zero relevant docs in top-10:

- ▶ HD-id: Flat **13%** → HNSW **71%**
- ▶ MM-id: Flat **78%** → HNSW **92%**
- ▶ Standard: within  $\pm 2$  pp on both datasets.

## Cost to match Flat recall (efSearch, HD):

- ▶ standard: **2.7 $\times$**  faster than Flat
- ▶ id: **40 $\times$**  slower than Flat

ANN's speedup evaporates on the population that needs it most.